

Rsearch

A user-friendly R interface to VSEARCH with integrated visualization and automated parameter tuning

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MOTIVATION

VSEARCH¹ is a fast, versatile open-source tool for metabarcoding analysis and a popular choice for high-diversity environmental DNA (**eDNA**) datasets, where it has been shown to outperform tools such as DADA2 and Deblur². However, its command-line interface may **limit accessibility** for many researchers. To address this, we developed **Rsearch**, an **R-based interface** to VSEARCH with integrated tools for **visualization** and **automated parameter tuning**.

Rsearch OVERVIEW

Rsearch supports **core metabarcoding workflow steps**, from preprocessing to clustering and taxonomic classification, and **extends VSEARCH** with visualization, parameter tuning, and data conversion tools.

KEY FEATURES

- Streamlined workflow in R
- Automated parameter tuning
- Integrated visualization
- Modular use of selected components
- Flexible data handling
- Easy integration with other R packages

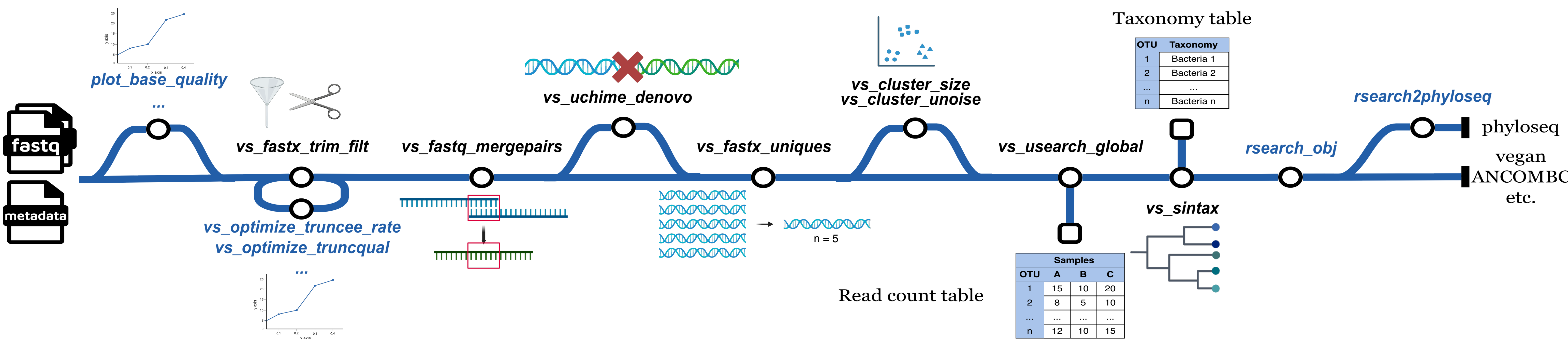


Figure 1. Example workflow for paired-end sequence analysis with *Rsearch*. The main path represents a typical metabarcoding pipeline, while detours indicate optional steps for visualization or automated parameter tuning.

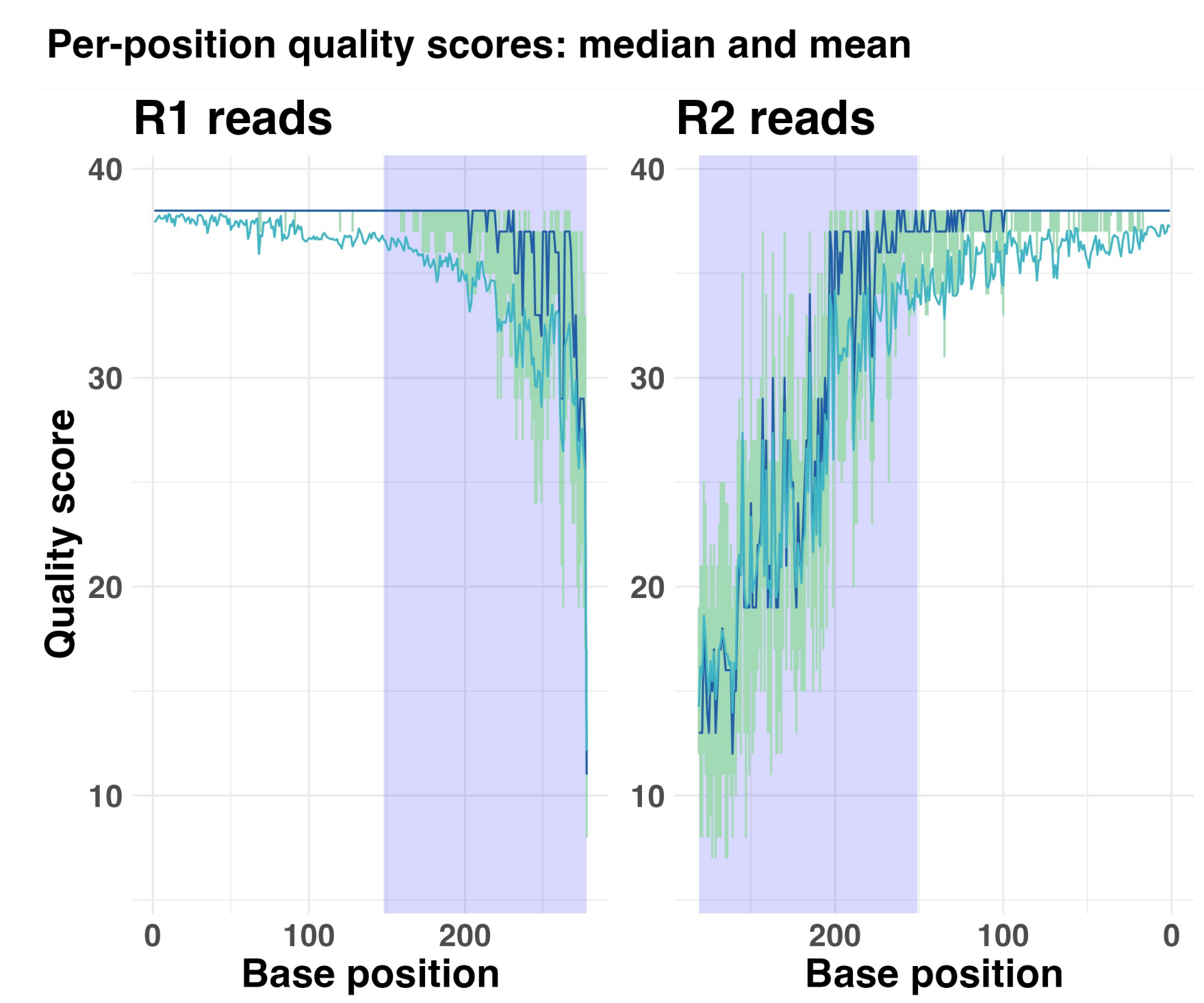


Figure 2. Inspect paired-end read quality with *plot_base_quality*. Shaded regions indicate read-pair overlap and help guide the 3' end trimming. This is a new feature of typical quality profile plots.

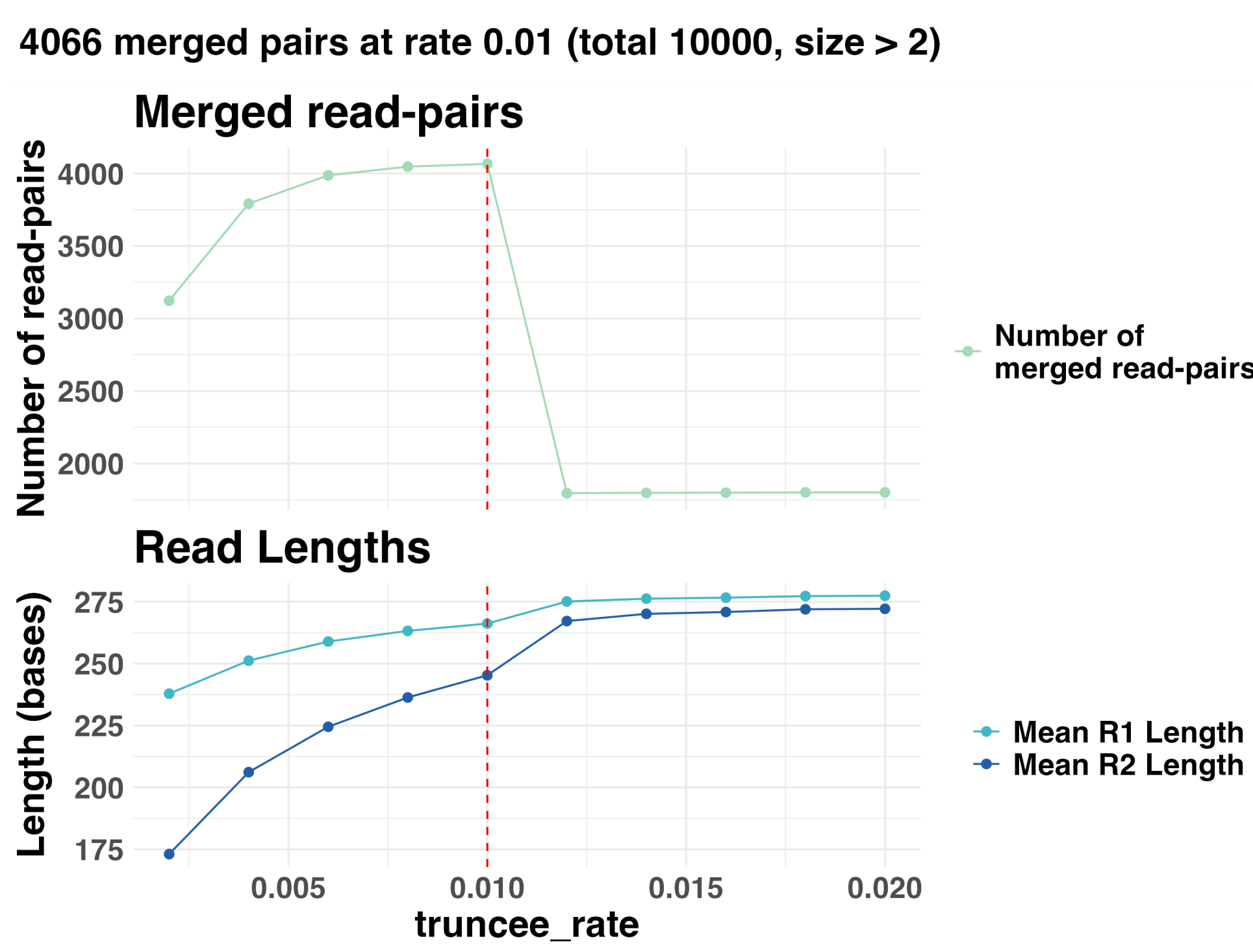


Figure 3. Automated tuning with *vs_optimize_trunce_rate*. *trunce_rate* truncates the 3' end of reads to keep the average expected error per base below a specified threshold. The red line marks the optimal value: the one that maximizes merged read pairs with copy number ≥ 2 , assuming sequencing errors are more common in singletons.

BENCHMARKING

Benchmarking against DADA2³ on a 16S rRNA mock community dataset showed that both pipelines produced **accurate abundance profiles** but differed in **diversity estimates**: DADA2 consistently over-split sequences, inflating low-prevalence OTUs, whereas **Rsearch** produced fewer, more prevalent, and more stable OTUs.

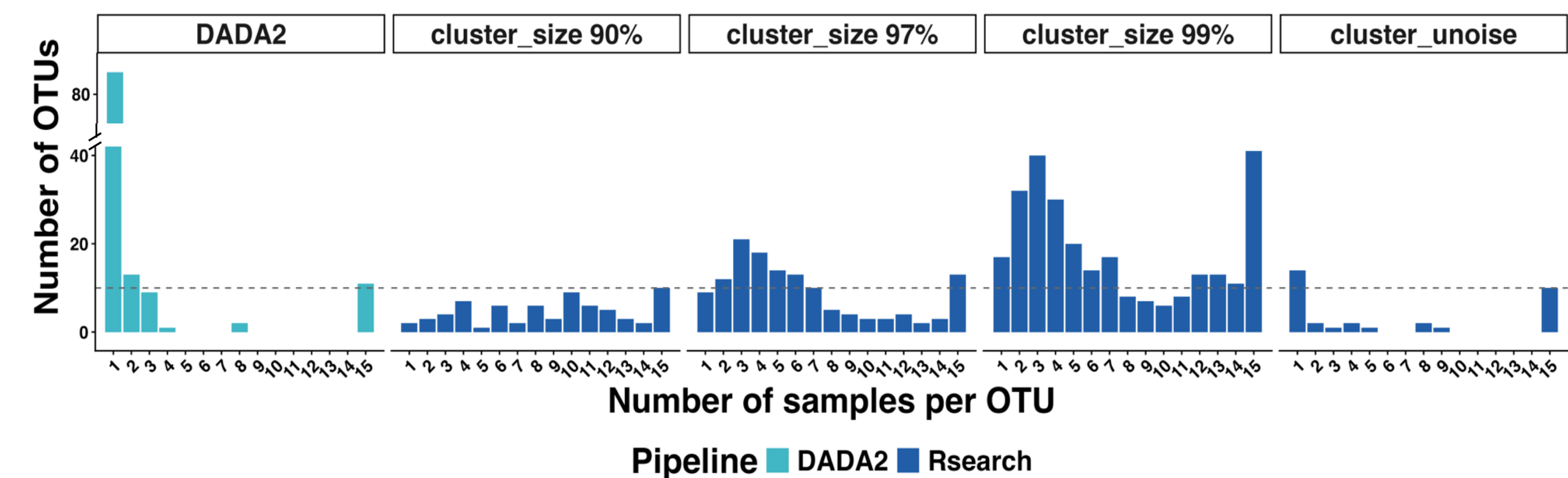


Figure 4. OTU sample prevalence. The horizontal dashed line indicates the expected 10 OTUs present across all 15 mock community samples.

CONCLUSIONS

Rsearch combines the efficiency of VSEARCH with the accessibility and flexibility of R, providing a reproducible and **user-friendly** framework for metabarcoding analysis. With integrated **visualization** and **automated parameter tuning**, it streamlines key workflow steps and supports analysis of **high-diversity datasets**.

QR code for CRAN

QR code for GitHub

QR code for Tutorial

Access Rsearch here

REFERENCES 1 Rognes, T., Flouri, T., Nichols, B., Quince, C. & Mahé, F. VSEARCH: a versatile open source tool for metagenomics. PeerJ 4, e2584 (2016). 2 Nilsen, T. et al. Swarm and UNOISE outperform DADA2 and Deblur for denoising high-diversity marine seafloor samples. ISME Commun. 4, ycae071 (2024). 3 Callahan, B. J. et al. DADA2: High-resolution sample inference from Illumina amplicon data. Nat Methods 13, 581–583 (2016).

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